LECTURE: FOURIER SINE SERIES

Today: We're *finally* going to solve the problem that's been bugging us for the last couple of lectures!

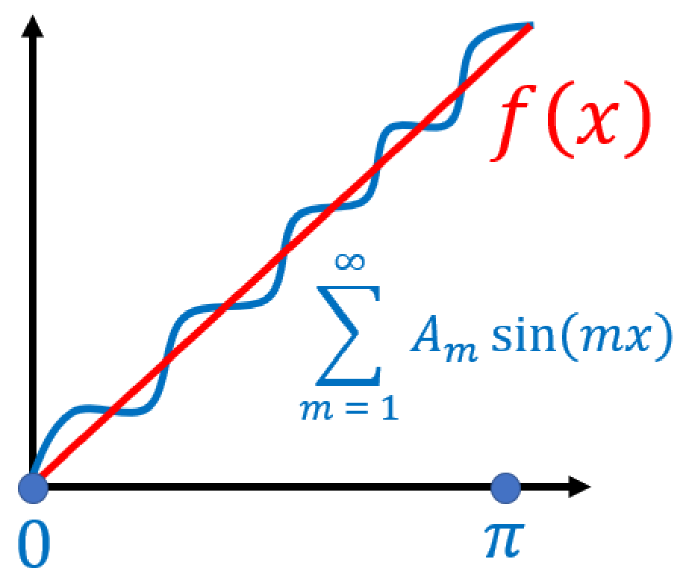
1. Prelude

Goal:

Given a function $f(x)$ on $(0, \pi)$ find A_m such that

$$f(x) = \sum_{m=1}^{\infty} A_m \sin(mx) \quad \text{Fourier Sine Series}$$

Intuitively: Want to write $f(x)$ in terms of sine waves



The key that allows us to solve this problem is some Linear Algebra!!

2. Orthogonality

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors

Definition:

$\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** (perpendicular) if

$$\mathbf{u} \cdot \mathbf{v} = 0$$

This gives us a 1 second way of checking for perpendicularity.

Definition:

$\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is **orthogonal** if $\mathbf{u} \cdot \mathbf{v} = 0$ and $\mathbf{u} \cdot \mathbf{w} = 0$ and $\mathbf{v} \cdot \mathbf{w} = 0$

That is, every vector is perpendicular to each other

We don't check $\mathbf{v} \cdot \mathbf{u}$ and $\mathbf{u} \cdot \mathbf{u}$ since $\mathbf{v} \cdot \mathbf{u} = \mathbf{u} \cdot \mathbf{v}$ and $\mathbf{u} \cdot \mathbf{u} > 0$ if $\mathbf{u} \neq \mathbf{0}$

Ultra Important Fact:

Suppose $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is an **orthogonal** set of **nonzero** vectors

Moreover, suppose $\mathbf{x} = a\mathbf{u} + b\mathbf{v} + c\mathbf{w}$ then we have

$$a = \frac{\mathbf{x} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \qquad b = \frac{\mathbf{x} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \qquad c = \frac{\mathbf{x} \cdot \mathbf{w}}{\mathbf{w} \cdot \mathbf{w}}$$

So if a set is orthogonal, then we get an *explicit* formula for a, b, and c

THIS is what makes orthogonality so important! In practice, it's a real pain to find a, b, c but if your set is orthogonal, you can find them quite easily.

Hugging Analogy: $\mathbf{x}$ hugs $\mathbf{u}$ ($\mathbf{x}\cdot\mathbf{u}$) and then $\mathbf{u}$ is so happy it hugs itself ($\mathbf{u}\cdot\mathbf{u}$) and same with $\mathbf{v}$ and $\mathbf{w}$

Why?

$$
\begin{aligned}
\mathbf{x}\cdot\mathbf{u} &= (a\mathbf{u}+b\mathbf{v}+c\mathbf{w})\cdot\mathbf{u} \\
&= a\,(\mathbf{u}\cdot\mathbf{u}) + b\cancelto{0}{(\mathbf{v}\cdot\mathbf{u})} + c\cancelto{0}{(\mathbf{w}\cdot\mathbf{u})} \qquad \text{(By orthogonality)} \\
&= a\ (\mathbf{u}\cdot\mathbf{u})
\end{aligned}
$$

Hence $a = \dfrac{\mathbf{x}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}$ and similar for b and c

3. Fourier Sine Series

What does this have to do with Fourier series?

Goal:

Given a (continuous) function $f(x)$ on $(0,\pi)$

$$\text{Can you find } A_m \text{ such that } f(x) = \sum_{m=1}^{\infty} A_m \sin(mx) \text{ on } (0,\pi)?$$

$\sum_{m=1}^{\infty} A_m \sin(mx)$ is called the **Fourier Sine Series** of $f(x)$ on $(0,\pi)$

We first need to define the dot product of two functions:

Definition:

Given functions $f(x)$ and $g(x)$ on $(0, \pi)$

$$f \cdot g = \int_0^\pi f(x)g(x)dx$$

Example 1:

If $f(x) = x^2$ and $g(x) = x^3$ then

$$f \cdot g = \int_0^\pi \left(x^2\right)\left(x^3\right) dx = \int_0^\pi x^5 dx = \frac{\pi^6}{6}$$

Note: Compare this with the usual dot product

$$(1,2,3) \cdot (4,5,6) = 1 \times 4 + 2 \times 5 + 3 \times 6 = 32 = \text{ Sum of Products}$$

$$\text{Similarly } \int_0^\pi \underbrace{f(x)g(x)}_{\text{Product}} dx = \text{ Sum of Products}$$

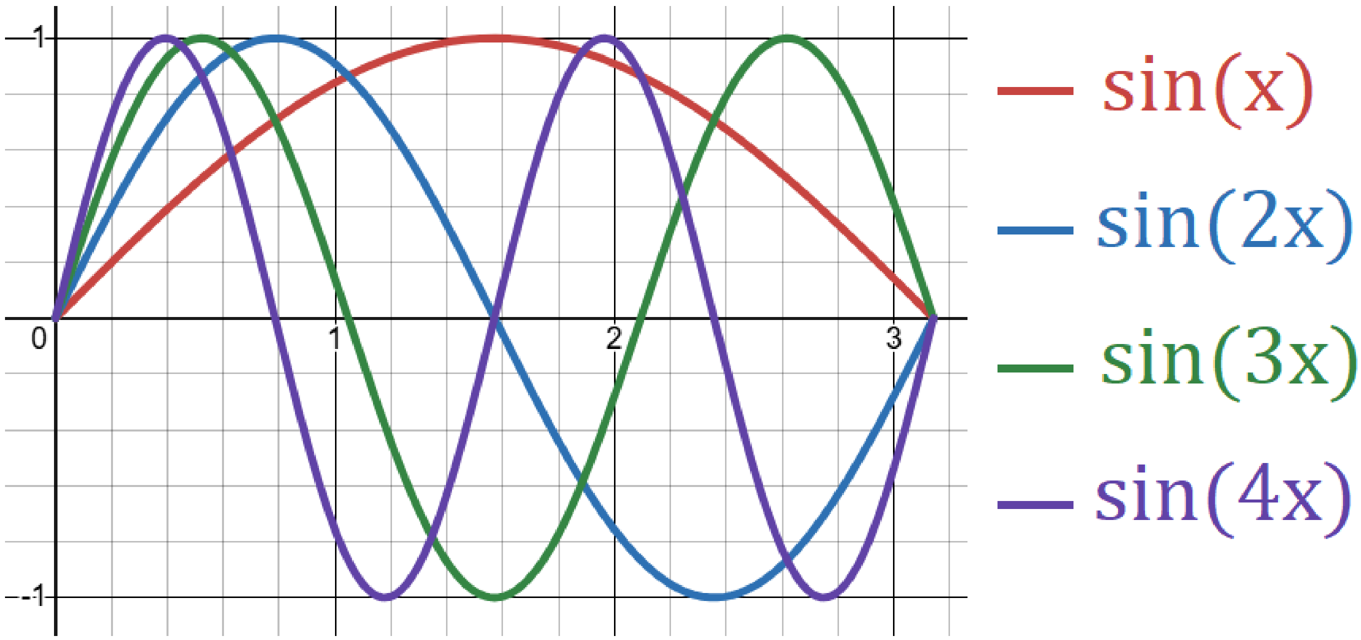
The key to solving our goal is that the $\sin(mx)$ in $\sum_{m=1}^\infty A_m \sin(mx)$ is orthogonal:

Fact:

The following is orthogonal on $(0, \pi)$ with the dot product above

$$\{\sin(mx) \mid m = 1, 2, \cdots\} = \{\sin(x), \sin(2x), \cdots\} \text{ orthogonal}$$

So all the $\sin(mx)$ functions in the following picture are perpendicular to each other **WOW**



Why? Show that for all $m \neq n$ we have

$$\sin(mx) \cdot \sin(nx) = 0 \text{ that is } \int_0^{\pi} \sin(mx)\sin(nx)dx = 0 \qquad \text{(See Homework)}$$

Consequence: In particular, if $f(x) = \sum_{m=1}^{\infty} A_m \sin(mx)$

Then from the Ultra Important above with $\mathbf{x} = f(x)$ and $\mathbf{u} = \sin(mx)$ and $a = A_m$

$$A_m = \frac{f(x) \cdot \sin(mx)}{\sin(mx) \cdot \sin(mx)} = \frac{\int_0^{\pi} f(x)\sin(mx)dx}{\int_0^{\pi} \sin(mx)\sin(mx)dx} = \frac{\int_0^{\pi} f(x)\sin(mx)dx}{\frac{\pi}{2}}$$

We used $\int_0^{\pi} \sin(mx)\sin(mx)dx = \int_0^{\pi} \sin^2(mx)dx = \frac{\pi}{2}$ (see Homework)

Note: Intuitively it's $\frac{\pi}{2}$ because $\frac{\pi}{2}$ is half the length of $(0, \pi)$

Summary: Fourier Sine Series

If $f(x) = \sum_{m=1}^{\infty} A_m \sin(mx)$ on $(0, \pi)$ then

$$A_m = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(mx) dx$$

You hug $f(x)$ with $\sin(mx)$, and $\frac{2}{\pi}$ comes from $\sin(mx)$ hugging itself

More generally, we have the following:

If $f(x) = \sum_{m=1}^{\infty} A_m \sin\left(\frac{\pi m x}{L}\right)$ on $(0, L)$ then

$$A_m = \frac{2}{L} \int_0^{L} f(x) \sin\left(\frac{\pi m x}{L}\right) dx$$

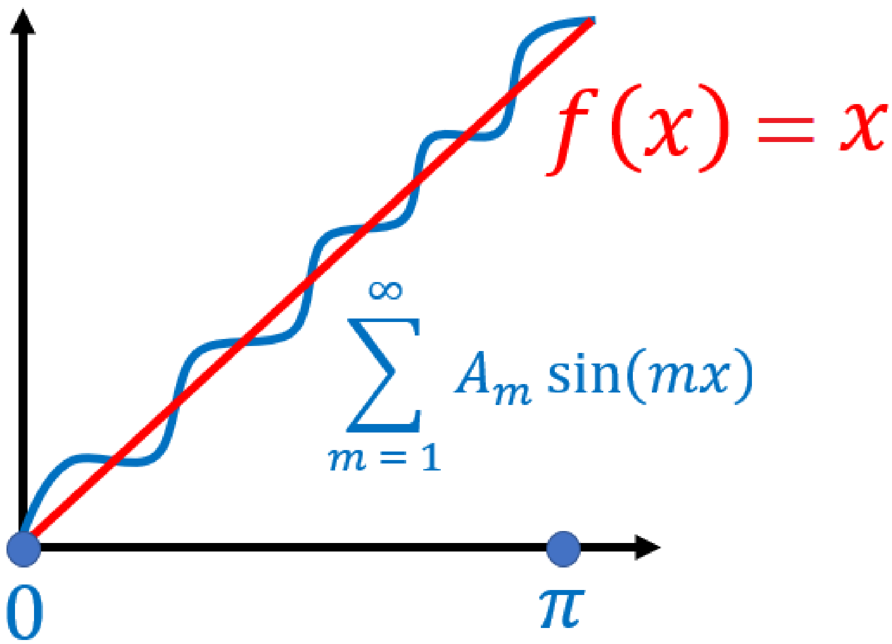
The $\sin\left(\frac{\pi m x}{L}\right)$ functions will naturally appear when we solve PDE

4. Example

Demo: Fourier series

Example 2:

$$\text{Find } A_m \text{ such that } x = \sum_{m=1}^{\infty} A_m \sin(mx) \quad \text{on } (0, \pi)$$



Using the formula from above with $f(x) = x$, we get

$$A_m = \frac{2}{\pi} \int_0^{\pi} x \sin(mx) dx$$

Now integrate by parts with $u = x$ and $dv = \sin(mx)$ we get

$$\begin{aligned}
A_m &= \frac{2}{\pi} \left(\left[x \left(\frac{-\cos(mx)}{m} \right) \right]_0^{\pi} - \int_0^{\pi} \left(\frac{-\cos(mx)}{m} \right) dx \right) \\
&= \frac{2}{\pi} \left(-\frac{\pi}{m} \cos(\pi m) + \frac{0}{m} \cos(0) + \frac{1}{m} \int_0^{\pi} \cos(mx) dx \right) \\
&= \frac{2}{\pi} \left(-\frac{\pi}{m} (-1)^m + \frac{1}{m} \cancelto{0}{\left[\frac{\sin(mx)}{m} \right]_0^{\pi}} \right) \\
&= \left(\frac{2}{m} \right) (-1)^{m+1}
\end{aligned}$$

Note: Here we used $\cos(\pi m) = (-1)^m$ which follows because

$\cos(\pi) = -1$ and $\cos(2\pi) = 1$ and $\cos(3\pi) = -1$ and $\cos(4\pi) = 1$ etc.

This tells us that

$$\begin{aligned} x =& \sum_{m=1}^{\infty} \left(\frac{2}{m}\right) (-1)^{m+1} \sin(mx) \\ =& \frac{2}{1}\sin(x) - \frac{2}{2}\sin(2x) + \frac{2}{3}\sin(3x) - \frac{2}{4}\sin(4x) + \cdots \end{aligned}$$

In the demo below, the Fourier series is getting closer and closer to x on $(0, \pi)$. The jumps will be explained in a future lecture:

Demo: Fourier series

Application 1: If you let $x = \frac{\pi}{2}$ in the above formula, then you get

$$\begin{aligned} \frac{\pi}{2} =& 2\sin\left(\frac{\pi}{2}\right) - \sin(\pi) + \frac{2}{3}\sin\left(\frac{3\pi}{2}\right) - \frac{1}{2}\sin(2\pi) + \frac{2}{5}\sin\left(\frac{5\pi}{2}\right)\cdots \\ \frac{\pi}{2} =& 2 - \frac{2}{3} + \frac{2}{5} - \frac{2}{7} + \cdots \end{aligned}$$

Dividing by 2 we get the following insane formula

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4} \qquad \text{WOW!!!}$$

Application 2: We can finally solve the heat equation problem that has been bugging us

Example 3:

$$\begin{cases} u_t &= Du_{xx} \\ u(0,t) &= 0 \\ u(\pi,t) &= 0 \\ u(x,0) &= x \end{cases}$$

PARTS I–V: From the Separation of Variables lecture we got

$$u(x,t) = \sum_{m=1}^{\infty} A_m e^{-m^2 Dt} \sin(mx)$$

PART VI: Initial Condition

$$u(x,0) = \sum_{m=1}^{\infty} A_m e^0 \sin(mx) = \sum_{m=1}^{\infty} A_m \sin(mx) = x$$

From the previous example, we obtained

$$A_m = \frac{2}{m}(-1)^{m+1}$$

PART VII: Solution

$$\begin{aligned} u(x,t) =& \sum_{m=1}^{\infty} \left(\frac{2}{m}\right)(-1)^{m+1} e^{-m^2 Dt} \sin(mx) \\ =& 2e^{-Dt}\sin(x) - e^{-4Dt}\sin(2x) + \frac{2}{3}e^{-9Dt}\sin(3x) - \cdots \end{aligned}$$

Demo: Heat Equation Example

The solution starts out as x but then quickly goes to 0 as $t \to \infty$